

SECTION 01 91 00

GENERAL COMMISSIONING REQUIREMENTS

SPECIFIER NOTE: A/E SHALL ADVISE UTSW PM AND CONFIRM ON EXTENT OF COMMISSIONING. A/E SHALL FULLY EDIT THIS SECTION FOR THE PROJECT SCOPE AND INCLUDE ADDITIONAL COMMISSIONING SECTIONS TO FULLY DOCUMENT SCOPE AND REQUIREMENTS.

PART 1 GENERAL

1.1 SUMMARY

- A. Commissioning (Cx) is a quality-oriented process of assuring that a facility is constructed and performs as intended in accordance with the contract documents.
 - 1. Confirm that all building systems (including building envelope) have been installed and exercised throughout their full range of intended operation, intended performance, and that anticipated failures have been initiated or simulated to verify response and recovery.
 - 2. Ensure all parties cooperate and participate throughout the construction process.
 - 3. Ensure that commissioning, including functional tests, O&M documentation review, and training is to occur after startup and initial checkout with deficiencies repaired and be completed before Substantial Completion
 - 4. Ensure that adequate time for all Cx activities is planned, developed, and communicated in a detailed Cx schedule. All members of Cx team shall provide input on the schedule.
 - 5. Verify that operation and maintenance manuals submitted to Owner are complete: Detailed operation and maintenance (O&M) data submittals by Contractor are utilized to achieve this.
 - 6. Verify that the Owner's operating personnel are adequately trained: Formal training conducted by Contractor is utilized to achieve this.

1.2 SCOPE OF COMMISSIONING

- A. Systems installed under this contract are to be inspected, tested, signed off as complete and operational, and operated for CxP verification as described in this specification.
- B. Systems to be commissioned by the CxP include the following:
 - 1. HVAC Systems: (Include Section 23 08 00 - Commissioning of HVAC Systems)
 - a. Major and minor equipment items.
 - b. Fire Smoke and Fire dampers
 - c. Piping systems and equipment.
 - d. Ductwork and accessories.
 - e. Terminal units.
 - f. Control system.
 - g. Sound control devices.
 - h. Vibration control devices.
 - i. Variable frequency drives.
 - 2. Special Ventilation / Equipment:
 - a. Laboratory Fume hoods:
 - b. Laboratory or Specialty pressurization.
 - c. Laboratory Exhaust.
 - d. Autoclaves.
 - e. Specialty fans.
 - f. Egress pressurization.
 - g. Animal Resource Center (ARC) Equipment: Bedding Vacuum System, Cage Wash, Ventilated Racks,

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- h. Animal Monitoring and Watering System (Refer to Section 13 31 13 - Animal Monitoring and Watering Systems).
- i. HEPA Filters: ISO 14644-3-2019, NIH, ASHRAE 52.1, 52.2.
- j. Dust Capture System.
- k. Gas Sensors.
- 3. Electrical Systems: (Include Section 26 08 00 - Commissioning of Electrical Systems).
 - a. Power quality.
 - b. Emergency power systems: (Refer to Section 26 32 13 - Standby Power Generator Systems)
 - c. Automatic Transfer Switches.
 - d. Uninterruptible power systems.
 - e. Lighting Controls other than manual switches.
- 4. Plumbing Systems: (Include Section 22 08 00 - Commissioning of Plumbing Systems)
 - a. Water heaters.
 - b. Booster pumps.
 - c. Laboratory gas systems.
 - d. Laboratory Compressed Air System.
 - e. Medical gas systems.
 - f. Landscape irrigation.
 - g. Emergency Fixtures.
 - h. Backflow preventers.
 - i. Deionized Water System.
 - j. Vacuum System - Waste and anesthetic gasses
- 5. Energy Management Systems.
- 6. Electronic Safety and Security:
 - a. Life Safety Systems.
 - b. Fire Alarm System.
 - c. Security system, including doors and hardware.
 - d. Access Control.
- 7. Fire Protection Systems (Include Section 21 08 00 - Commissioning of Fire Protection Systems).
 - a. Water-Based Fire Sprinkler System.
 - b. Pre-action Systems.
 - c. Clean Agent Fire Extinguishing System.
 - d. Kitchen Extinguishing System.
- 8. Integrated Automation.
- 9. Communications:
 - a. Voice and Data Systems.
 - b. Public Address / Paging.
 - c. Nurse Call / Hospital Communications.
- 10. Building Envelope:
 - a. Thermal and moisture integrity.
 - b. Air tightness.
- 11. Structural Systems.
- 12. Elevating and Conveying Systems.
- 13. Other equipment and systems explicitly identified elsewhere in Contract Documents as requiring additional requirements for commissioning.
 - a. JCAHO Regulations.
 - b. High Containment Spaces - Includes Bio Safety Level (BSL) 3 Laboratories. Follows CDC and NIH requirements, Third Party in compliance with Federal Select Agent Program (FSAP).
- 14. Sound Transmission Class-related interior partitions.
- 15. Indoor Air Quality Procedures:

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- a. The Commissioning Authority will coordinate, Contractor will execute; see Section 01 57 19 - Airborne Contaminants Control - Temporary Environmental Controls.
- C. The contractor shall be responsible for carrying out all work required for commissioning these systems that is defined as a contractor responsibility.

1.3 DEFINITIONS

- A. Commissioning Provider (CxP) - The Commissioning Provider has been contracted directly with the Owner for this project. The CxP has overall responsibility for planning and coordinating the commissioning process. However, commissioning involves all parties involved in design and construction process.
- B. Commissioning Coordinator (CxC) -- the individual appointed by the Contractor (and approved by the Owner) to act as the Contractor's single point-of-contact for all Cx related activities.
- C. Cx Team Members -- all parties who represent the Owner, A/E, Contractor, Subcontractors, Manufacturers, Vendors, and Suppliers and Consultants associated with the project.
- D. Contractor's Cx Record (CCR) -- the compilation of Cx related documentation including but not limited to the Commissioning Plan, the Equipment Matrix, Close-out Documentation Matrix, Cx Schedule, observation reports, inspections, meeting minutes, Cx checklists/testing scripts/procedures, etc.
- E. Building Automation System (BAS) – A control system designed to automatically control and monitor building systems.
- F. Owner's Project Requirements (OPR) – A formal document developed early in the design process that communicates what the Owner wants accomplished in the project.
- G. Basis of Design (BOD) – A formal document developed by the design team early in the design process that communicates how the systems designed will meet the project objectives.
- H. UW – University of Wisconsin
- I. AEE – The Association of Energy Engineers
- J. ACG – AABC Commissioning Group
- K. BCA – The Building Commissioning Association
- L. ASHRAE – The American Society of Heating, Refrigerating, and Air-conditioning Engineers

1.4 RELATED DOCUMENTS

- A. The Uniform General Conditions, applicable requirements of all Divisions of the Contract specifications and all Contract Drawings apply to work of this section. In the event of conflict between specific requirements of the various documents, the more restrictive, the more extensive (i.e., more expensive) requirement shall apply.

1.5 RELATED REQUIREMENTS

- A. Technical specifications throughout all Divisions of the Project Manual, which pertain to operable and non-operable equipment and/or building systems, are directly applicable to this section, and this section is directly applicable to them. Particular attention is directed to:
 - 1. Division 01 General Requirements, specifically
 - a. Section 01 32 00 – Project Planning and Scheduling
 - b. Section 01 45 00 – Project Quality Control
 - c. Section 01 77 00 - Closeout Procedures and Submittals
 - 2. Divisions 07-14
 - 3. Divisions 21-33 sections as appropriate, and Cx specific sections:
 - a. Section 21 08 00 - Commissioning of Fire Protection Systems
 - b. Section 22 08 00 - Commissioning of Plumbing Systems
 - c. Section 23 08 00 - Commissioning of HVAC Systems
 - d. Section 23 05 93 - Testing, Adjusting, and Balancing
 - e. Section 23 05 94 - System Testing, Adjusting and Balancing
 - f. Section 26 08 00 - Commissioning of Electrical Systems

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- g. Section 28 08 00 - Commissioning of Fire Alarm Systems
- h. Section 25 08 00 - Commissioning of Integrated Automation.

1.6 REFERENCES

- A. ASHRAE Standard 202 - 2018
- B. ASHRAE Commissioning Guideline 0 - 2019
- C. ASHRAE Guideline 1.1 - 2007
- D. Associated Air Balance Council Commissioning Guideline

1.7 COMMISSIONING TEAM AND RESPONSIBILITIES

- A. The commissioning team shall consist of representatives of the following:
 - 1. Owner (O)
 - a. Owner's Responsibilities
 - 1) Ensure the availability of the skilled and knowledgeable O&M staff for all scheduled instruction and demonstration sessions.
 - 2) Review the Commissioning Specifications and Checklists.
 - 3) Review all submittals pertaining to the contract documents and provide answers during the commissioning process.
 - 4) Participate, as appropriate, in onsite commissioning meetings.
 - 5) Owner may be on site to review commissioning documentation, to witness functional performance tests, and to analyze the installation and its performance.
 - (a) Successful demonstration of full systems integration in the facility shall be to the Owner's Designated Representative (ODF)
 - 6) Ensure the appropriate involvement of the Commissioning and project management Teams as required in the commissioning process.
 - 2. Design Team (A/E)
 - a. A/E Professionals maintain the responsibility of the final approval of the design and the installed systems.
 - b. Develop and provide the Owner Project Requirements (OPR) and Basis of Design (BOD) to the Commissioning Team.
 - c. Review all submittals pertaining to the Contract Documents and provide response to all questions during the commissioning process.
 - d. Review the Commissioning Specifications and Checklists.
 - e. Participate, as appropriate, in onsite commissioning meetings.
 - f. A/E may be on site to review commissioning documentation, to witness Functional Performance Tests, and to analyze the installation and its performance.
 - 3. Commissioning Provider (CxP)
 - a. The Cx Provider (CxP) shall be the leader of the commissioning team, planning all commissioning activities in coordination with the A/E, Contractor, Subcontractors, Third Party Testing Agencies, and Equipment Vendors.
 - b. The CxP shall witness and document the operation of the systems as conducted by the contractors.
 - c. The CxP shall provide the following:
 - 1) Perform a commissioning review of the OPR, BOD, and design documents prior to mid-construction documents and back-check the review comments in the subsequent design (and prior to construction commencing).
 - 2) Review contractor submittals applicable to the commissioned systems to ensure compliance with the OPR and BOD.
 - 3) Prepare commissioning plan and develop checklists for the commissioning process as specified herein.
 - 4) Prepare commissioning documents and ensure its distribution for review and comment.
 - 5) Revise the commissioning plan as required during construction.

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- 6) Coordinate commissioning meetings.
 - 7) Coordinate with the Contractor the commissioning activities among all contractors, sub-trades and suppliers.
 - 8) Monitor system verification checks and ensure the results are documented.
 - 9) Monitor controls point-to-point checks done by the controls contractor and ensure the results are documented.
 - 10) Observe equipment start-ups and initial system operations tests scheduled in coordination with the Contractor at the discretion of the CxP
 - 11) Attend and witness the contractors while operating equipment and systems as required to ensure that all required functional performance tests are carried out for verification purposes
 - 12) Witness all functional performance tests and document the results.
 - 13) Prepare and submit a Commissioning Report which documents all checks and tests done throughout the Commissioning process, and the results obtained from each.
 - 14) Ensure all required O&M manuals are provided.
 - 15) Ensure instructions and demonstrations (training) are provided to the Owner's designated O&M staff.
 - 16) Coordinate with O&M staff within 10 months after Substantial Completion a plan for resolving outstanding commissioning issues. Verify with UTSW during the 10 months duration.
4. General Contractor or Contractor (GC)
- a. Team Includes:
 - 1) Contractor Commissioning Coordinator (CxP)
 - 2) Subcontractors (Subs)
 - b. Contractor Team Responsibilities:
 - 1) Cooperate and coordinate work with the CxP and the commissioning process.
 - 2) Carry out all the physical activities required for installation and operation of components and systems during the commissioning process as required.
 - 3) Assign Contractor Commissioning Coordinator (CxP) as a full-time representative and primary point of contact to the Commissioning Team,
 - (a) Submit resume to Owner within 30 days after Notice to Proceed for review and approval. GC may hire a third party firm to perform the CxP role with Owner approval.
 - (b) CxP shall have Cx credentials from UW, BCA, AEE, ACG, or ASHRAE, or other Cx credentials can be submitted for evaluation and potential acceptability.
 - (c) CxP shall facilitate communications among all contractors and suppliers and other Commissioning Team Members to ensure the necessary cooperative action.
 - (d) CxP shall attend commissioning meetings and ensure action items are addressed as required to allow the commissioning process to precede on schedule.
 - 4) Ensure the cooperation and participation in the commissioning process of all Subcontractors and Vendors as applicable including meeting attendance.
 - (a) Assign a Subcontractor representative, as required and submit the person's name to the commissioning agency, within one (1) month of the award of the contract.
 - (1) Representative shall have the authority to make decisions on behalf of the Subcontractor as they relate to the organization and scheduling of commissioning activities.
 - (b) Each Subcontractor in this division shall include in their quotes the cost of participating in the commissioning process as specified herein including providing sufficient personnel during system verification and FPTs.
 - 5) Provide a complete set of as-built drawings and O&M manuals to the CxP for use during the project.

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- 6) Provide support and coordination on all interfaces for the building systems.
- 7) Develop and carry-out a comprehensive project schedule including inspections, O&M manual submission, training sessions, pipe and duct system testing, flushing and cleaning, equipment startup, testing and balancing, third-party testing and task completion for use by the CxP.
 - (a) Ensure commissioning is complete prior to substantial completion of the overall project.
 - (b) Schedule commissioning events to coincide with the completion of systems.
 - (c) Maintain the commissioning documents on site throughout the process.
- 8) Submit any Requests for Information (RFI) to the Design team, track and verify completion of each RFI.
- 9) Submittals:
 - (a) Generate and submit all submittals required to the required Owner, Architect, Engineers and the CxP.
 - (b) Include requirements for submittal data, start-up documentation for each piece of equipment in the project, alignment testing, vibration testing, operational testing, O&M data, and training.
 - (c) Notify the CxP when submittals are submitted, track all responses, and track resubmittals until all submittals are accepted.
 - (d) Submit finalized submittal matrix or checklist indicating all submittals have been accepted to the Owner and the CxP.
- 10) Ensure all systems and installations are prepared for commissioning testing.
 - (a) Carry out and document performance checks to ensure that all equipment and systems fully functional and ready for the CxP to witness formal Functional Performance Tests (FPTs).
 - (b) Ensure corrective action is taken and re-check until the results are satisfactory, and the system is ready for safe start-up.
 - (c) If during an attempted equipment or system start-up, deficient or incomplete work is discovered that would preclude safe operation, the start-up shall be aborted until corrective action has been taken. Ensure such action is taken and verified before re-scheduling a new start-up. Those responsible for deficient or incomplete work shall be responsible for costs.
 - (d) Operate equipment and systems for FPTs in accordance with the Commissioning Plan and as directed by the CxP. If improper functionality, incomplete work, or other deficiencies affecting system performance are discovered, the FPTs shall be stopped by the CxP. Those responsible for deficient or incomplete work shall be responsible for costs in accordance with this section. Ensure that all corrections necessary for full and complete system operation as specified are completed; applicable subcontractors carry out functional performance checks to confirm correct operation before applying to the CxP to reschedule the FPTs for the system in question.
 - (e) If improper functionality, incomplete work, or other deficiencies affecting system performance are discovered, the FPTs shall be stopped by the CxP. Those responsible for deficient or incomplete work shall be responsible for costs in accordance with this section.
 - (f) Ensure that all corrections necessary for full and complete system operation as specified are completed; applicable subcontractors carry out functional performance checks to confirm correct operation before applying to the CxP to reschedule the FPTs for the system in question.

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- 11) Notify the CxP a minimum of two weeks in advance of scheduled equipment and system start-ups, so that the CxP may witness system verifications, and equipment and system start-ups, if the CxP desires.
 - 12) Notify the CxP a minimum of two weeks in advance, of the time for start of the TAB work. Attend the initial TAB meeting for review of the TAB procedures. Cooperate with the TAB Firm.
 - 13) Generate the Owner Training Agenda and schedule in coordination with the Owner and CxP per Project requirements.
 - 14) Close-out Documentation:
 - (a) Gather O&M data on all equipment and assemble in binders as required by the Specifications. Submit to CxP prior to the completion of construction.
 - (b) Update drawings to as-built condition and review to verify all as-builts are accurate before submission.
 - 15) Provide instruction and demonstrations (training) for the Owner's designated operating staff, in conjunction with the participation of qualified technicians from major equipment suppliers and the controls contractor.
 - (a) Provide to the Owner and CxP all devices, such as portable operator's terminals and all software and hardware, for the owner representatives to have direct access to the entire systems to be used in completing the commissioning services. Verify the requirement to provide POT with software and hardware in the contract.
 - 16) In the event that any scheduled equipment or system start-ups or functional performance tests are terminated because the Owner, CxP or the A/E team discover deficient or incomplete work, or due to the non-attendance of required contractor or supplier personnel, the contractor or Subcontractor responsible for the termination shall also be responsible for paying reasonable costs of time and travel expenses of any or all of the following representatives who were physically present for the purpose of witnessing the start-up or the FPT: the CxP, A/E team and the Owner. The Owner may provide a statement to the Contractor identifying the specific activity that was terminated, the scheduled date, and a list of those in attendance, along with their reasonable time and travel expense costs.
5. Testing, Adjusting, and Balancing Firm (TAB)
- a. TAB Firm Responsibilities
 - 1) With respect to HVAC commissioning, the TAB firm shall:
 - (a) Include costs for HVAC commissioning requirements in the proposed price.
 - (b) Attend commissioning meetings scheduled by the CxP prior to, and during, on-site TAB work being done.
 - (c) Attend the TAB planning meeting scheduled by the Contractor. Be prepared to discuss the procedures that shall be followed in testing, adjusting, and balancing the HVAC system.
 - (d) At the completion of the TAB work, submit the final TAB report to the Owner, CxP and Design A/E Team.
6. Third Party Testing Agencies (TA)
7. Owner's Operations and Maintenance Staff (O&M)
- B. During the commissioning process, participation of team members shall generally be required as noted in the following sections of the Specification (with abbreviations as noted in brackets in the preceding list of team members).
- 1.8 WORK INCLUDED
- A. It is of primary concern that all systems installed in the project perform in accordance with the design intent and the OPR.

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- B. This section establishes minimum general and administrative requirements pertaining to Cx progress-tracking tools, documentation of installation, startup and performance testing of equipment, devices, assemblies and building systems.
 - 1. Additional technical and operational requirements for particular systems and components are established in the various technical sections of the specifications.
- C. The Contractor is solely responsible for the Cx process. This responsibility shall not be delegated to subcontractors, but by necessity will require the participation of subcontractors.
- D. UTSW PM will provide the Contractor with an electronic copy of standard Cx coordination and tracking tools, and document templates for various inspections, outage requests, reports, etc. Cx related documents, including tracking tools will be delivered to the Owner at Substantial Completion as part of the CCR.
 - 1. The Contractor shall customize the provided tools and templates to meet the specific details of the project.
 - 2. It is the responsibility of the Contractor to develop project-specific documentation forms and tracking tools where they are not provided by the UTSW PM
- E. The Owner's Quality Assurance (QA) testing and inspection program is independent of the Cx program.
 - 1. The Contractor is required to perform all testing (or provide or make available portions of the building for consultant and/or third party testing) as specified in the contract documents in addition to the Cx requirements outlined herein.
 - 2. Coordination of these processes by the Contractor will help minimize any duplication of effort.
 - 3. Unless stipulated elsewhere in the contract documents, QA testing and inspection documentation shall be included in the CCR.

PART 2 PRODUCTS

2.1 TEST EQUIPMENT

- A. Subcontractors shall provide all specialized tools, test equipment and instrumentation required to execute startup, checkout, functional performance, integrated systems and entire facility testing that includes equipment under their contract.
- B. Test equipment shall be of sufficient quality and degree of accuracy to test and/or measure system performance within tolerances specified.
 - 1. Subcontractors and vendors shall provide calibration certificates for all test equipment and instrumentation.
 - 2. A testing laboratory shall use test equipment that has been calibrated within the previous 12 months.
 - 3. Calibration shall be NIST traceable.
 - 4. Tested equipment shall be calibrated according to manufacturer's recommended intervals and recalibrated when dropped, damaged, or when Owner deems necessary.
 - 5. Calibration tags shall be affixed or certificates readily available.

PART 3 EXECUTION

3.1 COMMISSIONING PLAN

- A. The Commissioning Plan identifies processes and procedures necessary for a successful commissioning effort.
 - 1. The plan provides a step-by-step outline to assure the Owner's Project Requirements are met during the construction process.
 - 2. The plan also further defines the roles and responsibilities of all parties involved in the commissioning effort, including lines of communication and deficiency and conflict resolution.
 - 3. The Owner's CxP is responsible for the development of the Commissioning Plan.
 - 4. The Contractor is responsible for the implementation of the plan and applicable tools for the scheduling and tracking of the commissioning activities outlined in the plan.

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3.2 COMMISSIONING ACTIVITY DOCUMENTATION REQUIREMENTS

- A. System Verification Checklists (SVC) – Also known as Pre-Functional Checklists (PFC), SVCs are important to ensure that the equipment and systems are connected and operational and that Operational and Functional Performance Testing proceeds without unnecessary delays. The Contractor, appropriate Subcontractors and Vendors shall support the CxP in development of SVCs for each system and components by reviewing and commenting on the checklists in accordance with the Project Specifications. These checklists document the inspections and procedures necessary to take a piece of equipment from a static state into an operating state. These checklists augment the manufacturer's start-up checklists to provide a complete document from procurement to functional testing when combined. It is the responsibility of the Contractor and Subcontractors to complete the documentation of the checklists during construction.
- B. Functional Performance Tests (FPT) – The Owner's CxP shall develop the Functional Performance Test Forms in accordance with the Specifications and submit to the Contractor and appropriate Subcontractors for review and comment. The Contractor shall review the FPTs and submit to the Owner and A/E team for approval. Once approved, the forms shall be included in the Commissioning Plan for use during the commissioning process. The installing Contractors should use the approved FPTs as a basis for any pre-testing that shall occur prior to official Functional Performance Testing.
- C. Integrated Systems Testing (IST) – The IST documentation shall be used to document the demonstration of operational performance of multi-component sequences of operation to include integration with other components, interlocks, and alarm conditions for homogeneous systems. Testing procedures are to include all modes of operation and shall employ an itemized action-response format.
- D. Entire Facility Integration Test (EFIT) – The EFIT shall document demonstration of operational performance and integrated operation of the entire facility as a multi-functioning operational system. The Subcontractors shall operate each system, jointly and independently of other systems, through selected modes of operation including normal and emergency power modes and life safety interaction.
- E. The Owner's CxP will develop the EFIT test script in accordance with the Specifications and submit to the Contractor and Owner for review. The Contractor will review the EFIT and submit to the Owner and Design team for approval. Once approved, the test will be input into the Commissioning Plan for use during the commissioning process. The installing Subcontractors should use the approved EFIT as a basis for any pre-testing that will occur prior to official testing.
- F. The Contractor schedules the Entire Facility Integration Test and coordinates all involved members of the Commissioning Team. The Owner will coordinate any necessary personnel or facility impacts. Functional Performance Testing for all systems will be complete before scheduling the EFIT. The Contractor oversees the actual testing, which is usually performed by the installing Subcontractors or Vendors. The CxP will facilitate and guide the test utilizing the approved EFIT.

3.3 COMMERCIAL COMMISSIONING SOFTWARE SOLUTION

- A. The Contractor shall submit for approval by the Owner, a commercial Cx software solution used to generate and collect the CCR and associated Cx activity checklists and procedures, equipment information, associated manuals, photos, etc. in a database format that is COBie compliant, and that employs an issue/deficiency tracking system. The commercial Cx software solution shall provide for customizable systems and equipment types and designations such as, but not limited to, access control systems, security systems, building envelope systems and healthcare systems (nurse call, medical gas, pneumatic tube conveyance, etc.). The Cx Software shall:
 - 1. Provide for online storage of "library" files that can be organized in a customizable folder structure.
 - 2. Include sufficient licensing to accommodate the maximum users necessary to meet the needs of the project, including licenses necessary for Owner's project team. Coordinate with the Owner to determine license quantity requirements. Licenses and access to the Cx software will be required to be maintained for 12 months after Final Completion.
 - 3. Employ security capabilities using industry standard encryption (128-bit minimum) for web-based access and offline device synchronization. The software solution shall employ the use of a unique

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user ID and password for each individual user. Access and permissions shall be assignable to each unique user ID, or to categories or groupings of users engaged in similar roles. The Cx software shall allow for simultaneous multiple user access to database records for checklist updates, entry of issues, attachment of photos, access to library files, etc.

4. Allow for custom data elements (attributes) associated with each type of equipment (VAV Box, Air Handling Units, Electrical Panels, etc.) that can be configured by the user. Each project must be able to support a unique (user defined) set of data elements specific to the requirements of the project. The system shall provide support for industry standard barcode or QR code for each equipment/system item.
5. Provide for checklist generation (creation) with customizable checklist responses. Software shall employ the use of checklist templates to create individual equipment-specific checklists. Software shall provide for auto "Issue Generation" based on user-selected trigger responses. Responses shall include a default "Pass/Fail", numeric (only) response and user-defined "single item select list" as a minimum.
6. Provide for tracking of systems/equipment status as each item progresses through the Cx process. The process (progressive list of statuses) shall be user defined and selectable from a drop-down or "single item select list."
7. Provide for interactive field data entry in either an online or offline environment. The data entry or Cx software remote (field) access device shall be Android and/or iOS compatible. Contractor shall provide (and maintain) Owner with Android or iOS compatible remote access devices for the duration of the project. (Examples are an iPad or tablet PC.) Offline access capability shall support storage of all database items enabling full software functionality. The remote access device shall be capable of storing for offline access contract documents (drawings, specifications, etc.) and software library documents.
8. Provide browser-based access to the online database via the internet using current industry standard browsers (e.g., Chrome, Fire Fox, or Safari). Browser-based access must be fully compatible with a minimum of two of the listed industry standard browsers.
9. The use of any software that does not comply with each of the requirements listed above or use of an alternative methodology of generating/collecting/documenting the CCR shall require a written substitution proposal that includes samples of each tracking tool and document type (checklist, report, equipment tracking, etc.) that will be included in the CCR. Substitution proposals must list each criterion in Section 3.1 and designate compliance or detail specific non-compliance. Substitution proposal shall be submitted prior to Contractor's Notice to Proceed (NTP) for project construction phase.

3.4 COMMISSIONING SCHEDULE

- A. The objective of scheduling Cx process activities is to integrate and coordinate Cx activities with other construction phase activities. Detailed scheduling will allow Cx Team members to coordinate work with other team members in order to complete all Cx activities prior to Substantial Completion. The Cx Schedule shall include major Cx activities, essential prerequisites for major equipment and system activities and operable equipment/system/assembly functional and integrated systems performance demonstrations. The Cx Schedule shall account for Test and Balance (TAB) consultant activities properly scheduled and coordinated into the project work flow to allow for the completion of all TAB work prior to Substantial Completion. The Cx Schedule shall account for building envelope consultant inspections, tests and other activities properly scheduled and coordinated into the project work flow to allow for inspections and testing prior to covering or concealment. As applicable, the following milestones and activities shall be incorporated into the project master construction schedule:
 1. Cx Kickoff Meeting
 2. Building Automation System Submittal Approval
 3. Control Sequence of Operation Coordination Meeting (reference Specification Section 23 05 93 Testing, Adjusting, and Balancing)

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4. Ethernet Connectivity
 5. Building Envelope Testing and Verification Activities
 6. Major HVAC Equipment/Systems Startup
 7. System Specific Test and Balance Activities
 8. Major HVAC Equipment Functional Performance Tests
 9. Integrated System Tests
 10. Entire Facility Integration Test
 11. Training
- B. The Contractor shall provide a minimum of 72 hours' notice prior to each Cx activity to the Owner and the CxA and Testing Consultants when applicable, unless stipulated otherwise in this specification or other technical specifications.

3.5 COMMISSIONING KICKOFF MEETING

- A. Within 120 days after the effective date of the NTP for the construction scope or package that includes building operational systems, the CxP will schedule a date to conduct a Cx kickoff meeting with all parties involved in the Cx process. At a minimum the meeting should include the major subcontractors, specialty manufacturers/suppliers, the A/E, mechanical and electrical consultants, the Owner's testing, adjusting and balancing (TAB) firm, the CxC, the Owner and representatives from the UT institution.
1. The CxP shall prepare for the meeting by creating drafts of the following documents for review at the meeting:
 - a. The Cx Plan
 - b. Equipment List
 - c. Closeout Documentation Matrix
 - d. The Cx Record - Table of Contents
 - e. An overview or demonstration of the approved Cx software
 - f. Sample ESCs and FPTs
 - g. Preliminary Cx Schedule incorporating Cx activities to coincide with the work flow contained in the master construction schedule
 2. The CxP shall conduct the meeting and review the Cx process and specifications, including discussion of documentation requirements, available test procedures and Cx forms.
 3. The Cx Plan review shall outline roles and responsibilities of each Cx team entity and the potential schedule impact as related to Cx requirements.
 4. The CCR Table of Contents review shall include discussion of the scope of work. The Contractor shall be prepared to distribute copies of the pertinent document samples to the subcontractors involved in the Cx process.
 5. The Cx Schedule review shall include the dates and durations for major systems startup, and shall identify functional performance testing that is included in the master construction schedule. Team members should identify potential schedule impact pertaining to their scope of work and test sequencing.
 6. Commissioning shall be an agenda item for project progress meetings until separate Cx meetings are deemed necessary.

3.6 PRE-INSTALLATION MEETINGS

- A. At a minimum, the Contractor shall schedule a separate meeting for the work involving each major building system or systems and assemblies. The pre-installation meeting shall be scheduled, in writing, a minimum of five (5) days in advance, and shall be scheduled so that the CxP, the Owner, Owner-designated consultants and Architect/Engineer can attend, if necessary. The meeting shall be convened following approval of system submittals and prior to commencement of system installation work.
1. The Contractor shall arrange for all subcontractors, suppliers and manufacturers involved in the system to be present or adequately represented.
 2. The Contractor shall bring the following to this conference, at a minimum, for review and discussion:
 - a. The portion of the equipment list applicable to the system/trade under discussion

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- b. Current work schedule data pertaining to the equipment delivery, installation, required testing, construction checklists, equipment startup and functional performance testing anticipated
- c. Copy of all approved submittals for the system
- d. Draft of documentation/checklists to be used for inspection, startup and functional performance testing of the system(s)/assembly under review.
- e. The purpose of this meeting is for the Contractor and all applicable subcontractors and/or suppliers and/or factory representatives to discuss all aspects of the installation of the particular system, testing and documentation required and procedures to be followed. Special attention is to be directed to the scheduled order of work and any impact on or by any other building systems.

3.7 CONTRACTOR'S VERIFICATION OF INSTALLATION – CONSTRUCTION CHECKLIST

- A. The Contractor shall document using a construction checklist a review of testing/inspection forms to ensure each is accurate and complete. This documentation shall include, but not be limited to, first-hand knowledge of the following items:
 - 1. Equipment/system is delivered in new condition and in accordance with approved submittals. Delivered equipment and materials are protected, staged, and stored in accordance with the specifications and the manufacturer recommendations.
 - 2. Each component device has been installed and terminated in accordance with the project specifications and governing codes as well as the manufacturer's written recommendations.
 - 3. All shop drawings and product data submittals have been approved for each component device.
 - 4. All valve schedules, wiring diagrams, control schematics, electrical panel directories, etc. have been submitted, approved, and equipment/systems installed in accordance with specifications.
 - 5. All test reports and/or certifications required have been submitted and accepted. If required, certificate of acceptance from manufacturer representative and/or engineering technician have been received. Provide copies of all checklist/inspection documentation completed by the manufacturer or certified technician.
- B. The Contractor shall be responsible for correction of all noted deficiencies. Any request for inspection/re-inspection or test/retest of a device or system shall first be confirmed as being compliant by the Contractor before submitting a request to the Owner for inspection/re-inspection and testing/retesting.

3.8 EQUIPMENT STARTUP

- A. Startup of Independent Systems, Assemblies, Components and Devices
 - 1. Equipment startup is a documented formal Cx activity for the CxP and the Owner (and others) to verify and witness proper startup in accordance manufacturer recommendations and contract documents. The Contractor shall utilize the Equipment Startup Checklist (ESC) to document the activity, participants, and witnesses.
 - 2. The Contractor shall not energize or activate, or allow activation of any operable device prior to equipment startup by a manufacturer representative. It is NOT permissible to "bump" motors prior to equipment startup. Contractor shall verify proper electrical service wiring (phasing) with the use of a phase rotation meter.
 - 3. The Contractor and manufacturer's representative shall inspect and accept the installation and preparedness for startup. The Contractor shall execute startup under supervision of a responsible manufacturer's representative in accordance with manufacturer's instructions and as specified in the contract documents. The installation shall not vary from provisions of the applicable specifications and the manufacturer's written recommendations for startup. The Contractor shall develop and use the ESC to document (for the CCR) preparedness for startup, startup procedures and record operational measurements and data appropriate for the equipment and in accordance with the technical specifications. The Contractor is encouraged to incorporate any manufacturer provided installation and startup checklists as part of the ESC.
 - 4. The Contractor shall provide five (5) business days' notification of scheduled equipment startup to the Owner, the Owner's designated consultants and the A/E team.

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3.9 FUNCTIONAL PERFORMANCE TESTING

- A. After all relevant Construction Checklists are completed and startup has been accomplished, the Contractor shall coordinate pre-functional testing by the subcontractors and applicable Owner consultants in preparation for the Functional Performance Test (FPT). Contractor shall provide five (5) business days' notice of the scheduled FPT (demonstration) to the CxP, the Owner, Owner designated consultants and A/E team.
- B. The Demonstration of an FPT is a documented formal Cx activity for the CxP and the Owner (and others) to verify the operation of equipment/assembly/system in accordance with contract documents to include all modes of operation, sequences of operation and anticipated failure conditions. The Contractor shall operate, or cause to be operated, each system, device, assembly, or equipment item, both intermittently and continuously, for the duration indicated in the specification section(s) for such item and/or in accordance with the manufacturer's written recommendations, and in accordance with the approved FPT procedures. The details of these activities shall be documented for the CCR. The Owner and/or an Owner-designated consultant shall witness and verify the results of the functional performance test demonstration.
- C. For operable equipment/systems, each component device and each building system shall be exercised to the full extent of its capability, from minimum to maximum, under automatic and manual control, and in bypass when applicable. The equipment/assembly/system shall be exercised using the Campus BAS graphics on a campus operator's workstation. All inputs, outputs and calculated values, as displayed on the operator's workstation graphics, shall be verified and documented.
- D. The Contractor and, when applicable, manufacturer's representative, shall supervise and coordinate adjustments, alignments, calibrations and balancing of all devices, equipment and systems for proper operation as part of the pre-functional testing activities.
- E. The Contractor shall coordinate with the Owner's consultants to support the progression and completion of their scope of work. The Contractor shall provide the TAB firm and envelope consultant with installation and performance data as requested by the consultant (and approved by the Owner).
- F. For Static systems, periodic observations shall be documented in accordance with manufacturer installation guidelines and recommendations. Performance testing as recommended by the manufacturer and in accordance with applicable technical specifications requirements shall be documented.
- G. Where final TAB of a system or particular components thereof are not specifically indicated to be performed by Owner or Owner's consultants, the Contractor is to provide final balancing and adjustments for operation within specified tolerances and provide documentation of it prior to scheduled FPT of each system.

3.10 INTEGRATED SYSTEM TESTING

- A. After successful completion and documentation of all system/assembly/equipment FPTs, the CxP shall schedule a meeting with the Project Cx Team to review the approved Integrated System Tests (ISTs) and demonstration procedures for each designated system. An integrated system contains two or more system components that have been functionally tested and have physical, hardwired or software interfaces that require one component to respond as the result of the operation of one or more other components. Examples of a few systems that would require an IST are: chilled water distribution, domestic hot water system, primary electrical distribution, fire alarm, access control and security systems.
- B. Development of IST and demonstration procedures may vary with each project. The CxP is responsible for providing ISTs that include all modes of operation for the system that could act upon or react to operation of separate system components. An IST shall be submitted to the Owner for review and comment and final approval by the A/E team. Collaboration with the A/E design team during IST development will ensure that a thorough performance demonstration is achieved. ISTs shall include a comprehensive, action-response checklist for all modes of operation and failure conditions included in the sequence of operation and shall itemize for each action, the anticipated response from each integrated system and/or associated component.

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- C. Following compliance with the provisions noted above and following submission of Operating & Maintenance (O&M) Manuals for the all systems to be demonstrated, the Contractor shall provide the Owner a five (5) business day notice of their intent to perform an IST demonstration. The CxP is responsible for documenting the results of the ISTs.

3.11 ENTIRE FACILITY INTEGRATION TESTING

- A. After successful completion and documentation of all ISTs, the CxP shall schedule a meeting with the Project Cx Team to review the Entire Facility Integration Test (EFIT). The EFIT is a facility-wide test to verify that all building systems interact and predictably perform in accordance with the design documents.
- B. Development of the EFIT and demonstration procedures shall be a collaborative effort of the Cx Team facilitated by the Contractor. The Contractor is responsible for ensuring that all building systems are included in the EFIT, that each system responds to designed modes of operation, and that anticipated failure conditions are itemized for monitoring and verification. The format of the EFIT testing procedures shall be an action-response matrix that identifies for each action (mode of operation), the itemized list of responses that are to be verified and documented. The CxP shall submit the EFIT to the Owner and A/E team for review and comment.
- C. Following compliance with the provisions noted above, the Contractor shall provide the Owner a five (5) business day notice of their intent to perform an EFIT. The CxP is responsible for documenting the results of the EFIT.

3.12 BUILDING ENVELOPE COMMISSIONING

- A. General: Comply with the following procedural requirements:
 - 1. NEBB S110 Whole Building Technical Commissioning of New Construction.
 - 2. ASTM E779 Standard Test Method for Determining Air Leakage Rate by Fan Pressurization.
 - 3. ASTM E1827 Standard Test Methods for Determining Airtightness of Buildings Using an Orifice Blower Door.
 - 4. ANSI/RESNET/ICC 301 Standard for the Calculation and Labeling of the Energy Performance of Low-Rise Residential Buildings using an Energy Rating Index.
 - 5. ANSI/RESNET/ICC 380 Standard for Testing Airtightness of Building Enclosures, Airtightness of Heating and Cooling Air Distribution Systems, and Airflow of Mechanical Ventilation Systems.
- B. Verify that the building envelope has been sufficiently completed for testing to commence.
- C. Conduct ongoing inspections as construction progresses to document satisfactory installation conditions related to thermal and moisture integrity of the building envelope that become concealed upon completion of construction.
- D. Submit a detailed narrative of proposed pressure test procedures prior to the test. Include a plan view showing proposed installation locations (personnel doors or other similar openings) for blower doors (or flexible ducts for trailer-mounted fans, if used).
- E. Avoid testing on days forecast to experience high winds, rain, or snow.
- F. Test the completed building and demonstrate that the air leakage rate of the building envelope does not exceed the specified requirements.
 - 1. Use equipment and methods necessary to produce indoor/outdoor pressure differential of 0.2 inches w.g. (50 Pascals).
 - 2. Use RESNET accredited software to document salient features of project's building envelope and to calculate the HERS Index Score.
- G. Determine location and nature of undesirable air leakage pathways using methods specified in ASTM E1186-17 Standard Practices for Air Leakage Site Detection in Building Envelopes and Air Barrier Systems.
- H. Deficiencies: Correct deficiencies and re-inspect or re-test, as applicable, at no extra cost to Owner.
 - 1. If difficulty in correction would delay progress, report deficiency to the Commissioning Authority immediately.

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2. Insulation for remedying building envelope deficiencies evidenced as excessive air leakage is specified in Section 07 21 00 - Thermal Insulation
3. Air barriers for remedying building envelope deficiencies evidenced as excessive air leakage are specified in Section 07 27 00 - Air Barriers.
4. Sealants for remedying building envelope deficiencies evidenced as excessive air leakage are specified in Section 07 92 00 - Joint Sealants

3.13 FIELD TESTING AND COMMISSIONING OF PARTITIONS FOR NOISE ISOLATION

- A. Conduct testing of partitions requiring a specific STC class indicated on drawings and/or in various specifications sections. Comply with ASTM E336 for testing methods, including requirements of Annex A1 for reduction of flanking sound transmission.
- B. Confirm that the FSTC values are not less than 67 percent of design STC values.
- C. Deficiencies: Correct deficiencies and re-inspect or re-test, as applicable, at no extra cost to Owner.
 1. If difficulty in correction would delay progress, report deficiency to the Commissioning Authority immediately.
 2. Sealants for remedying flanking sound transmission deficiencies evidenced as excessive air leakage are specified in Section 07 92 00 - Joint Sealants.

3.14 OWNER TRAINING

- A. Refer to Section 01 79 00 - Demonstration and Training.
- B. Training shall consist of classroom type sessions and on-site demonstrations of system operation. See specification technical sections for specific system/equipment requirements. If a system/equipment requires both field demonstration and training, they may be combined if the Owner approves.
- C. The Contractor shall provide a professional-grade video recording of training, with clear and intelligible audio, in accordance with the technical specifications. The Owner will select those portions of the training to be recorded.
- D. The Contractor shall be responsible for submitting individual training plans and for coordination, scheduling and completion of the training for all equipment as specified in the contract documents. The training will be conducted by the installing subcontractor and/or manufacturer's representative for each specific piece of equipment in accordance with the applicable technical specification sections. Each training plan shall be submitted to the Owner no later than 14 calendar days in advance of proposed training. Training plans shall include the specifications section reference, proposed trainer, and relevant qualifications (resume), training agenda with learning objectives, copies of training materials/handouts/visual aids, training date, time, location and duration.
- E. Training shall use the O&M Manuals as a basis for instructing the Owner's personnel regarding system operation. Training shall include a review of the contents of O&M Manuals and a review of equipment data and performance verification from the FPT checklists.
- F. Demonstrate in the field: startup, operation, control, adjustment, trouble-shooting, servicing, maintenance, each component device and shutdown of the system(s).
- G. Demonstrate both in the field and with the use of operator (workstation) graphics a detailed check-out at each stage of the sequences of operation. All equipment graphics, alarms and sequences of operation are to be reviewed, and demonstrated to the extent the Owner agrees is feasible.
- H. The Contractor shall participate in demonstration of Owner Furnished/Contractor Installed equipment in accordance with applicable technical specifications.
- I. As a minimum, the Contractor shall perform training on all Life Safety systems including, but not limited to, the following (if system is part of the project):
 1. HVAC and Controls
 2. Fire Alarm
 3. Fire Sprinkler Systems (including pumps)
 4. Elevator/Escalator
 5. Smoke Purge

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6. Stairwell Pressurization
7. Communications Systems
8. Emergency Power/Generator/UPS
- 9.
10. Alternative Energy/Energy Recovery Systems
11. Facility Security System
12. Medical Gas Systems
13. Security/CCTV/Access Control/Article Protection Systems

3.15 OPERATION AND MAINTENANCE MANUALS

- A. See Section 01 78 00 - Closeout Submittals for additional requirements.
- B. Add design intent documentation furnished by Architect to manuals prior to submission to Owner.
- C. Submit manuals related to items that were commissioned to Commissioning Provider for review; make changes recommended by Commissioning Provider.
- D. Commissioning Authority will add commissioning records to manuals after submission to Owner.

END OF SECTION

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